When Does Contiguity Help Watermark Detection? Sharp Phase Diagrams for Segment-Surviving Gumbel-max Watermarks

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Abstract

A Gumbel-max watermark is detected from one "pivot" per token, which is uniform on text the watermarked model did not produce and pushed toward 1 on text it did. After human editing, only a small fraction of tokens keep the watermark. The existing optimal theory treats these survivors as scattered independently, and its tests ignore token order. In practice, when a person rewrites some sentences and keeps others, or pastes AI passages into their own writing, the survivors form contiguous segments. We keep the same null hypothesis but let the alternative be "the watermark survives in segments of at least ℓ tokens." In an explicit theoretical sandbox we show: (i) tests that ignore token order are stuck at the old detection boundary however long the segments are; (ii) contiguity helps exactly when one segment carries enough evidence, measured by its expected log-likelihood ratio (for nearly deterministic text, the number of non-greedy word choices it contains); (iii) at the critical scale, Higher Criticism applied across blocks of tokens attains a sharp boundary, which reaches full detectability exactly when one segment's expected log-likelihood ratio exceeds $\log N$, the log of the number of blocks. A key-only count test captures most of the gain, and simulations agree with the theory.

1 Introduction

Two strands of the course meet in this project. The first is Higher Criticism (HC) and the phase-diagram theory of rare, weak signals (Lectures 2–3): when a few of many test statistics carry a small signal, HC detects them up to an explicit boundary in the (rarity, strength) plane. The second is LLM watermarking (Lecture 5), in particular the Gumbel-max watermark, whose per-token p-values can be combined by a sum or by HC. Li, Ruan, Wang, Long and Su (2025b) joined the two. When a human edits watermarked text, only a fraction ϵ of tokens keep the watermark; modeling each token as surviving independently, they derived the detection boundary and showed that an HC-type test attains it.

The question. Survival is rarely independent across tokens. Suppose a student asks a model to rewrite an essay and then keeps a few of the AI sentences verbatim, or pastes AI

paragraphs into their own document. The tokens that still carry the watermark then form contiguous *segments*: whole sentences or passages. The tests in the i.i.d. theory (the sum test, HC, and the truncated goodness-of-fit test of Li et al.) depend only on the histogram of pivot values, not on where the large ones occur, so they cannot see this structure. Windowed detectors exist in practice, but there is no theory of when, and by how much, looking at token order helps. We ask:

When the surviving watermark forms segments, how does the detection boundary change, and which tests attain it?

Our formulation. The null hypothesis is the one from class: no token carries the watermark. Only the alternative changes, from "each token survives independently with probability ϵ" to "the surviving tokens form runs of length at least ℓ, covering at least an ϵ fraction of the text." A test must work for every such arrangement of runs. The hardest arrangement turns out to be Li et al.'s model one level up: tile the text into blocks of ℓ tokens (think of sentences) and let each block survive independently. The sparsity is then over blocks rather than tokens, which suggests summing the evidence within each block and applying HC across blocks, exactly as in Lecture 3 but with blocks as the units.

The main insight: contiguity is worth as much evidence as a segment carries. A watermarked token carries evidence only to the extent that the model had freedom in choosing it. If the model was certain of the next token, the watermarked choice is the forced one and its pivot is exactly uniform. The more freedom, the more the key steers the choice and the further the pivot is pushed toward 1; formally, a token's expected evidence is $\mathrm{KL}(F_{P_t}\|\mathrm{Unif})$ (Section 4). When the model is nearly deterministic, this evidence arrives in rare bursts: with probability about Δ_t the model departs from its top choice, and only then does the pivot land extremely close to 1, which we call a *watermark event*. The signal is then sparse at two levels, across tokens (only a fraction ϵ survive) and within surviving tokens (only a fraction Δ produce events). Grouping tokens into a segment helps only if the segment accumulates enough evidence: enough events, or more generally a large enough expected log-likelihood ratio.

Results. Within an explicit sandbox (Section 3) we prove:

(i) **Order-blind barrier** (Proposition 5.1). Any test that ignores token order has exactly the i.i.d. detection boundary, whatever the segment length.

(ii) **Coarse phase diagram** (Theorem 5.2, Figure 1). With $\epsilon = n^{-p}$, $\Delta = n^{-q}$ and segment length $\ell = n^{\gamma}$ ($\gamma \neq q$), detection is possible if and only if $2p + q < 1$ (the i.i.d. region) or $\gamma > q$ (segments containing many events).

(iii) **Sharp boundary at the critical scale** (Theorems 5.3–5.4). When one segment's expected log-likelihood ratio is of order $\log n$, the boundary is a variational formula in the segment's cumulant, attained by HC across blocks. Its two ends are universal: it starts strictly above the i.i.d. boundary, and reaches full detectability exactly when a segment's expected log-likelihood ratio is at least $\log N$.

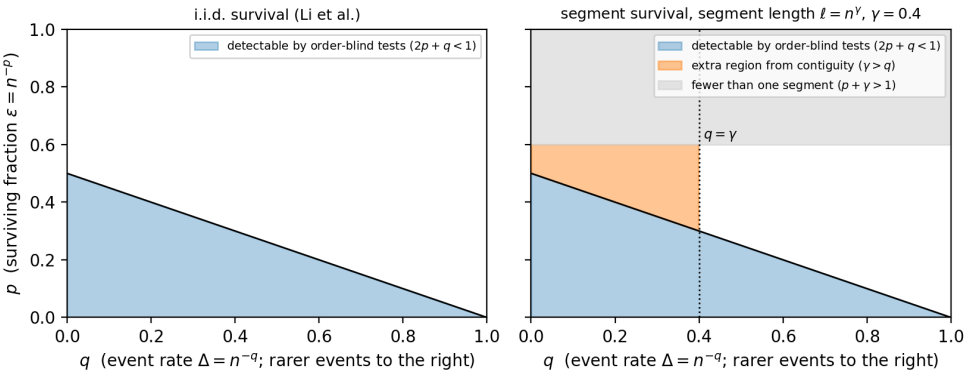


Figure 1: Coarse phase diagram (Theorem 5.2). Left: under i.i.d. survival, detection is possible iff $2p+q<1$ (Li et al.). Right: under segment survival with segments of length n^{γ}, segment-aware tests also detect in the orange region, where segments are longer than $1/\Delta$ ($q<\gamma$). Order-blind tests remain confined to the blue region.

(iv) **A key-only test** (Corollary 5.5). Counting large pivots per block and applying HC across blocks needs only the key, has its own sharp boundary, and recovers most of the gain.

Simulations (Section 6) confirm the predictions, including outside the sandbox (unknown segment positions and lengths, heterogeneous next-token distributions).

What is new. Li et al.'s boundary $2p+q=1$ already reflects that Δ acts like a sparsity parameter rather than a signal strength, and the general boundary we use for sparse mixtures (Theorem 5.3) is equivalent to a result of Cai and Wu (2014), which we reprove in a form convenient for blocks of tokens. What is new is the analysis of *contiguously* surviving watermarks: the evidence-per-segment criterion, the proof that the i.i.d.-optimal order-blind tests are strictly suboptimal under contiguous survival, and sharp boundaries for both an oracle test and a key-only test.

2 Background

2.1 The Gumbel-max watermark (review)

At step t an LLM has a next-token distribution (NTP) $P_t = (P_{t,w})_{w\in\mathcal{W}}$ over the vocabulary $\mathcal{W}$. In the Gumbel-max watermark (Aaronson, 2023; Kuditipudi et al., 2024), a secret key and the recent context produce pseudorandom numbers $U_{t,w} \overset{\text{iid}}{\sim} \text{Unif}(0,1)$, one per vocabulary item, and the model emits

$$w_t = \operatorname*{argmax}_{w\in\mathcal{W}} U_{t,w}^{1/P_{t,w}} = \operatorname*{argmax}_{w\in\mathcal{W}} \big\{ \log P_{t,w} + G_{t,w} \big\}, \qquad G_{t,w} = -\log(-\log U_{t,w}).$$

The $G_{t,w}$ are standard Gumbel, so by the Gumbel-max trick w_t is an exact draw from P_t: averaged over keys, watermarked text has exactly the model's distribution. The key holder recomputes the uniform number attached to the emitted token, the *pivot* $Y_t = U_{t,w_t}$.

Lemma 2.1 (Pivot law; Li et al. 2025a)**.** *If token t was not produced by the watermarked sampler (e.g., a human wrote it), then $Y_t \sim \mathrm{Unif}(0,1)$. If it was, then $Y_t \overset{d}{=} V^{P_{t,W}}$ with $W \sim P_t$ and $V \sim \mathrm{Unif}(0,1)$ independent; hence*

$$F_{P_t}(y) \equiv \mathbb{P}(Y_t \le y) = \sum_w P_{t,w}\, y^{1/P_{t,w}}, \qquad f_{P_t}(y) = \sum_w y^{1/P_{t,w}-1}, \qquad y \in [0,1].$$

Intuition. The key is independent of anything a human writes, so a human-chosen token's pivot is exactly uniform; the null needs no model of human writing. For a watermarked token, the pivot is a uniform number raised to the power $P_{t,W}$, the probability of the chosen token. A token chosen with certainty ($P_{t,W} = 1$) has a uniform pivot; a long shot ($P_{t,W}$ small) wins only when its key value was extreme, so its pivot is very close to 1. (Proof in Appendix B.)

Detectors from class, and freedom Δ_t. The class detectors use the scores $S_t = -\log(1-Y_t) \sim \mathrm{Exp}(1)$ or the p-values $1 - Y_t \sim \mathrm{Unif}(0,1)$ under the null, and reject for a large sum $\sum_t S_t$ or a large HC of the p-values. Both are *order-blind*: unchanged if the tokens are shuffled. Let $\Delta_t = 1 - \max_w P_{t,w}$, the probability that sampling departs from greedy decoding at step t (near 0 for "The capital of France is", perhaps 0.5 for a content word in open-ended prose), and let $\mathcal{P}_\Delta = \{P : \max_w P_w \le 1-\Delta\}$. Li et al. (2025a) show that the two-point law

$$P^\Delta = (1-\Delta,\ \Delta), \qquad f_\Delta(y) = y^\alpha + y^\beta, \quad \alpha = \tfrac{\Delta}{1-\Delta},\ \beta = \tfrac{1}{\Delta} - 1,$$

is least favorable in $\mathcal{P}_\Delta$ for their efficiency criterion when $\Delta \le 1/2$; we use it as the canonical hard case.

2.2 Detection under edits as a sparse mixture (Li et al.)

Li et al. (2025b) let each token independently carry the watermark with probability ϵ_n, so the pivots form a sparse mixture,

$$H_0:\ Y_t \overset{\text{iid}}{\sim} \mathrm{Unif}(0,1) \qquad \text{vs.} \qquad H_1:\ Y_t \sim (1-\epsilon_n)\mathrm{Unif}(0,1) + \epsilon_n F_{P_t}, \quad P_t \in \mathcal{P}_{\Delta_n},$$

the rare/weak setting of Lecture 3. With $\epsilon_n = n^{-p}$ and $\Delta_n = n^{-q}$, detection is possible if $q + 2p < 1$ and impossible if $q + 2p > 1$, and a truncated goodness-of-fit (Tr-GoF) test, a family of HC-like divergences between the empirical CDF of $1 - Y_t$ and the uniform CDF, attains the boundary adaptively. Sum-based tests reach only $q + p < 1/2$.
Intuition. Count pivots above $1 - \Delta$. The null count is $n\Delta \pm \sqrt{n\Delta}$, and the watermark adds about $n\epsilon\Delta$ (Lemma B.1). Detection needs $n\epsilon\Delta \gg \sqrt{n\Delta}$, i.e., $n\epsilon^2\Delta \to \infty$, which is $2p + q < 1$.

2.3 Related work and tools

Detecting signals on unknown intervals is classical: scan statistics maximize a local statistic over windows, Arias-Castro, Donoho and Huo (2005) derived optimal thresholds for such geometric signals, and Cai, Jeng and Li (2012) studied many short segments in genomics. In

watermarking, WinMax (Kirchenbauer et al., 2024) scans windows for pasted watermarked passages; WaterSeeker (Pan et al., 2025) and adaptive segmentation tests (Li, Liu and Li, 2025) localize watermarked segments; Xie et al. (2025) design watermarks to localize post-generation edits; Kuditipudi et al.'s (2024) edit-distance detector aligns text with the key sequence to tolerate insertions and deletions; and Li et al. (2025c) estimate the watermarked proportion under the i.i.d. mixture. None of these gives detection boundaries for contiguously surviving Gumbel-max watermarks or compares with the i.i.d.-optimal order-blind tests. Our proofs use the standard tools behind Lecture 3's "undetectable" region, which we stated but did not prove in class: second-moment and Hellinger lower bounds (Ingster, 1997; Donoho and Jin, 2004), Cai and Wu's (2014) boundary for general sparse mixtures, and large deviations on the $\log n$ scale (Dembo and Zeitouni, 1998). Appendix A reviews them.

3 Problem formulation and sandbox

All results are proved in the following idealized model; for each assumption we say what it means, where it comes from, and what it leaves out.

Data and survivor set. The final (edited) text has n tokens with NTPs $P_1, \dots, P_n$, and the detector, holding the key, computes the pivots $Y_1, \dots, Y_n$. Let $S \subseteq [n]$ be the unknown set of positions that still carry the watermark.

Assumption A1 (Pivot model; from Li et al.)**.** Given S and the NTPs, the pivots are independent with $Y_t \sim F_{P_t}$ for $t \in S$ and $Y_t \sim \mathrm{Unif}(0,1)$ for $t \notin S$. Write $\mathbb{P}_S$ for their joint law.

Meaning. The standard idealization of the statistical watermark literature: NTPs fixed and keys fresh at every step. *It ignores* that P_t depends on the edited context, that keys repeat when contexts repeat, and any evidence in the text itself.

Assumption A2 (Segment survival; new)**.** We test $H_0 : S = \emptyset$ against $H_1 : S \in \mathcal{S}_n(\epsilon, \ell)$, where

$$\mathcal{S}_n(\epsilon,\ell) = \Big\{ S = \textstyle\bigcup_j I_j : \ I_j \text{ disjoint intervals, } |I_j| \geq \ell, \ \sum_j |I_j| \geq \epsilon n \Big\}.$$

Meaning. The watermark survives on at least ϵn tokens, in runs of at least ℓ tokens; the number, positions and lengths of the runs are unspecified, so tests must work for every arrangement and need no model of the editor. The single parameter ℓ controls contiguity: $\ell = 1$ allows any set (the i.i.d. setting), and larger ℓ rules out fragmented survival. Runs arise from span-level editing and copy-paste: with context hashing, a token keeps the watermark only if it and its h predecessors are unedited, so rewriting whole sentences leaves the kept sentences as runs, while a pasted passage of length ℓ_0 yields a run of length $\ell_0 - h$. Scattered token edits do *not* create runs: if each token is kept independently with probability ϵ_0, a survivor's successor survives with probability ϵ_0, so in the sparse regime (ϵ_0 small) runs have mean length $1/(1-\epsilon_0) \approx 1$. Our lower bounds show that the hardest member of the class is *block-i.i.d. survival*: tile the text into n/ℓ blocks ("sentences") and keep each independently,

which is Li et al.'s model one level up. *It ignores* partial survival inside a segment and any distribution over segment lengths.

Assumption A3 (NTP regularity; from Li et al.)**.** For $t \in S$, $P_t \in \mathcal{P}_\Delta$; the coarse results hold uniformly over this class. The sharp results instead fix the NTPs (Assumption A6); the main examples are a fixed NTP P and the two-point law P^Δ.

Assumption A4 (Asymptotic calibration; as in Lecture 3)**.** $\epsilon_n = n^{-p}$ with $p \in (0,1)$; write $L = \log n$. In the *coarse regime*, $\Delta_n = n^{-q}$ with $q \in (0,1)$ and $\ell_n = n^\gamma$ with $p + \gamma < 1$. In the *critical regime*, one segment's expected log-likelihood ratio is of order $\log n$: either $\Delta_n = n^{-q}$ and $\ell_n \Delta_n = \kappa \log n$, or the NTPs are fixed and $\ell_n = c \log n$. There are then $N = \lfloor n/\ell_n \rfloor = n^{a+o(1)}$ blocks, with $a = 1 - q$ or $a = 1$ respectively, and we take $p < a$ (many segments). In these units Li et al.'s boundary $2p + q = 1$ reads $p = a/2$.

Tests and optimality. A test ψ is any function of the pivots, judged by its worst-case risk $R_n(\psi) = \mathbb{P}_\emptyset(\psi = 1) + \sup_{H_1} \mathbb{P}_S(\psi = 0)$, the supremum over all S and NTPs allowed by the assumptions. The problem is *detectable* if some test has $R_n \to 0$ and *undetectable* if every test has $R_n \to 1$. An *oracle* test may use the NTPs; a *key-only* test uses only the pivots. Table 1 summarizes the sandbox.

In the sandbox	Outside the sandbox
independent pivots, fixed NTPs (A1)	NTPs depending on edited context; repeated keys
runs of length $\geq \ell$ covering $\geq \epsilon n$ tokens (A2)	the editing process; partial survival in a run; localization
Δ-regular NTPs (A3); same NTPs in every block for sharp results (A6)	NTPs varying across blocks (probed by E4)
power-law calibrations, $n \to \infty$ (A4)	finite-n constants; calibration at tiny false-positive rates
pivot-only tests	tests that also use the text's likelihood

Table 1: The theoretical sandbox at a glance.

4 Evidence per token and per segment

For an NTP P, let $m_P(\theta) = \mathbb{E}_{\mathrm{Unif}}[f_P(Y)^\theta]$ be the moment generating function of a token's log-likelihood ratio $\log f_P(Y)$ under the null (for a non-degenerate NTP with finite support, finite exactly for $\theta > -P_{\max}/(1 - P_{\max})$). Three of its values carry meaning:

$$\log m_P(0) = \log m_P(1) = 0, \qquad (\log m_P)'(1) = \mathrm{KL}(F_P \| \mathrm{Unif}), \qquad \log m_P(2) = \log\big(1 + \chi^2(F_P \| \mathrm{Unif})\big).$$

The KL is the expected evidence of one watermarked token, and χ^2 is its second-moment information. Since pivots are independent, a block of watermarked tokens has null cumulant $\sum_t \log m_{P_t}(\theta)$ and expected log-likelihood ratio $\sum_t \mathrm{KL}(F_{P_t}\|\mathrm{Unif})$: *evidence adds over tokens.* When the model is nearly deterministic, this evidence takes a universal form.

Lemma 4.1 (Per-token evidence for small freedom)**.** *Let $m_\Delta = m_{P^\Delta}$ and $\varphi(\theta) = \int_0^1 \frac{(1+w)^\theta - 1}{w}\, dw - \theta$. Uniformly over compact sets of $\theta \in \mathbb{R}$,*

$$m_\Delta(\theta) = 1 + \Delta\, \varphi(\theta) + O(\Delta^2).$$

Hence $\chi^2(F_\Delta\|\mathrm{Unif}) = \Delta/2 + O(\Delta^2)$ and $\mathrm{KL}(F_\Delta\|\mathrm{Unif}) = \Delta\varphi'(1) + o(\Delta)$. The function φ is smooth and strictly convex, with $\varphi(0) = \varphi(1) = 0$, $\varphi(2) = \frac{1}{2}$, $\varphi'(1) = \frac{\pi^2}{12} + 2\log 2 - 2 \approx 0.209$ and $\varphi'(2) = \frac{\pi^2}{12} + 4\log 2 - \frac{11}{4} \approx 0.845$.

Intuition. For small Δ, the likelihood ratio $f_\Delta(y)$ is about $1 + e^{-u}$ at $y = 1 - u\Delta$ and about 1 elsewhere: the watermark is a small bump that at most doubles the density of pivots just below 1, coming from the rare events in which the model departs from its top choice. Two consequences. First, the information per token is *linear* in Δ, the signature of a rare, strong signal (a weak Gaussian mean shift δ has information of order δ^2). Second, a block of ℓ tokens has cumulant $\ell\Delta\,\varphi(\theta)$: it depends on the NTPs only through $\ell\Delta$, the expected number of *events* per segment. Since $\mathrm{KL} \approx 0.21\,\Delta$, "events per segment" is the small-freedom reading of "evidence per segment." For moderate freedom the picture changes: even the top token, if chosen with probability 0.5, has pivot $V^{0.5}$ with mean 2/3, so every watermarked token carries some evidence and the KL, not the event count, is the right measure.

5 Main results

5.1 Order-blind tests cannot use contiguity

Proposition 5.1 (Order-blind barrier)**.** *Let $P_t \equiv P_n$ for all t, and suppose $n\epsilon_n^2\chi^2(F_{P_n}\|\mathrm{Unif}) \to 0$ (for P^{Δ_n} this is $2p+q>1$; for a fixed P it is $p > 1/2$; in critical units both read $p > a/2$). Every test invariant under permutations of $(Y_1, \ldots, Y_n)$ has $R_n \to 1$, for every segment length $\ell_n \le \epsilon_n n$.*

Intuition. An order-blind test sees only the histogram of the pivots, which shuffling leaves unchanged. So it performs on any arrangement of runs exactly as on a uniformly random set of the same size, which is the i.i.d. problem, where Li et al.'s lower bound applies. (Second-moment proof in Appendix C.)

5.2 The coarse phase diagram

Theorem 5.2 (Coarse phase diagram)**.** *In the coarse regime ($\ell_n = n^\gamma$, $p + \gamma < 1$):*

(i) If $\ell_n\Delta_n = O(1)$ (in particular if $\gamma < q$) and $2p + q > 1$, the problem is undetectable.

(ii) If $2p + q < 1$, the global count test, which rejects when $\#\{t : Y_t > 1 - \Delta_n\} > n\Delta_n + (n\Delta_n)^{1/2}L$, has $R_n \to 0$.

(iii) If $\gamma > q$, the block scan has $R_n \to 0$. It tiles $[n]$ by blocks of length $b = \lfloor \ell_n/2 \rfloor$, counts $C_k = \#\{t \in \text{block } k : Y_t > 1 - \Delta_n\}$, and rejects if $\max_k C_k > \lambda + \sqrt{3\lambda L}$, where $\lambda = b\Delta_n$.

Parts (ii) and (iii) hold uniformly over $\mathcal{P}_\Delta$. Hence, for $\gamma \neq q$, the detectable region is $\{2p + q < 1\} \cup \{\gamma > q\}$ (Figure 1); on $\{2p + q > 1,\ \gamma > q\}$ order-blind tests fail while the block scan succeeds.

Intuition. Segments shorter than $1/\Delta$ rarely contain even one event, so contiguity gives nothing to aggregate and the i.i.d. boundary stays. Segments with polynomially many events are each individually conspicuous, so a scan finds one as soon as any exist. Without knowing (Δ, ℓ), dyadic grids of thresholds and block lengths with a Bonferroni correction give the same conclusions (Appendix C). The case $\gamma = q$ is the critical regime, next.

5.3 The sharp boundary at the critical scale

The coarse diagram jumps from "contiguity is useless" to "any segment can be found." The transition happens when one segment's expected log-likelihood ratio is of order $\log n$: enough to be suggestive but not individually conclusive, just like the $\sqrt{2r \log p}$ signals of Lecture 3, whose log-likelihood ratio $r \log p$ is also of order $\log p$.

Reduction to a sparse mixture of blocks. Tile the text into N blocks aligned with the segments. Each block is all-null or all-watermarked, and its evidence is summarized by its log-likelihood ratio T. The problem becomes rare/weak detection over N "super-observations," a fraction ϵ of which carry signal, with a non-Gaussian law. Abstractly: there are N independent blocks with law Q_n under the null and G_n if watermarked, H_0 : no watermarked block, H_1 : at least ϵN. Let $T = \log \frac{dG_n}{dQ_n}$ and $\Lambda_n(\theta) = \log \mathbb{E}_{Q_n} e^{\theta T}$.

Assumption A5 (Block cumulants)**.** $N = n^{a+o(1)}$ with $a \in (0, 1]$, $\epsilon = n^{-p+o(1)}$ with $0 < p < a$, and $\Lambda_n(\theta)/L \to K(\theta)$ for every θ in an interval (θ_-, ∞) with $\theta_- \in [-\infty, 0)$. On this interval K is finite, differentiable and strictly convex, and, if $\theta_- > -\infty$, $K(\theta) \to \infty$ as $\theta \downarrow \theta_-$. Let $t_+ = \lim_{\theta \to \infty} K'(\theta) \in (K'(1), \infty]$ (finite, e.g., when per-token log-likelihood ratios are bounded, as for a fixed NTP).

Then $K(0) = K(1) = 0$. Let $t_0 = K'(0) < 0 < t_1 = K'(1)$ and $I(t) = \sup_\theta\{\theta t - K(\theta)\}$; then $I(t) - t$ is the corresponding rate under G_n, and $I = +\infty$ on (t_+, ∞). Define

$$D(K, a) = \frac{a}{2} + \sup_{t \geq t_0} \Big[t - I(t) + \tfrac{1}{2} \min\{I(t), a\} \Big].$$

The test ψ^* (HC across blocks). Let $\pi_k = Q_n(T > T_k) + V_k\, Q_n(T = T_k)$, with $V_k \overset{\text{iid}}{\sim} \text{Unif}(0, 1)$, be randomized block p-values, exactly uniform under H_0 even for discrete T (unrandomized discrete p-values would bias HC's numerator). Let $\text{HC}_N^{\text{dy}} = \max_{1 \leq j \leq \log_2 N} \big(\#\{k : \pi_k \leq 2^{-j}\} - N2^{-j}\big)/\sqrt{N2^{-j}}$ be HC at dyadic levels. The test ψ^* rejects if $\text{HC}_N^{\text{dy}} > L^2$ or $\min_k \pi_k < (NL)^{-1}$ (a Bonferroni component). It needs the null block law Q_n but not (p, ϵ).

Theorem 5.3 (Sharp boundary for sparse block mixtures)**.** *Under Assumption A5: if $p > D(K, a)$ the problem is undetectable, and if $p < D(K, a)$ then $R_n(\psi^*) \to 0$.*

Intuition. Count blocks whose evidence exceeds tL. By large deviations the null count has mean $n^{a-I(t)}$, while the watermark adds about $n^{a-p-(I(t)-t)}$ such blocks. The count succeeds if the excess beats both 1 and the null standard deviation $n^{(a-I(t))/2}$; optimizing over t gives $p < D$. HC searches all thresholds at once, and the Bonferroni component covers thresholds where the null count is below 1. The matching lower bound is a Hellinger bound with Chernoff tails, and a convex-duality argument shows the two exponents coincide (Appendix D). The condition $p < D$ is Cai and Wu's (2014) boundary in different units, and for Gaussian blocks it reduces to the Donoho–Jin boundary of Lecture 3 (Remark D.1).

Application to watermarks. By Section 4, a watermarked block's cumulant is the sum of its tokens' cumulants. The sharp results assume this sum is the same for every block.

Assumption A6 (Block-homogeneous NTPs)**.** Segments are aligned to a grid of $N = n^{a+o(1)}$ blocks of length ℓ_n. Every block has the same multiset of NTPs, known to the detector (an oracle), and the block cumulant $\Lambda_n(\theta) = \sum_{t \in \text{block}} \log m_{P_t}(\theta)$ satisfies Assumption A5 with limit K.

Meaning. Blocks are interchangeable, so the abstract theorem applies. Two instances: *fixed NTPs*, $P_t \equiv P$ and $\ell_n = c \log n$, giving $a = 1$ and $K = c \log m_P$; and *small freedom*, $P_t \equiv P^{\Delta_n}$ with $\ell_n \Delta_n = \kappa \log n$, giving $a = 1 - q$ and $K = \kappa\varphi$ (Lemma 4.1). *It ignores* NTPs varying across blocks, where a worst-case alternative would put the segments on the least informative stretches (code, boilerplate).

Theorem 5.4 (Sharp boundaries for segment-surviving watermarks)**.** *Under Assumption A6, with limiting block cumulant K:*

(a) ***Optimal boundary.*** *If $p < D(K, a)$, then ψ^* applied to the block log-likelihood ratios $T_k = \sum_{t \in \text{block } k} \log f_{P_t}(Y_t)$ has $R_n \to 0$. If $p > D(K, a)$, the problem is undetectable.*

(b) ***Saturation.*** *$D(K, a) \geq a$, i.e., every configuration with $p < a$ is detectable, if and only if $K'(1) \geq a$. Here $K'(1)L = (1 + o(1)) \sum_{t \in block} \mathrm{KL}(F_{P_t} \| \mathrm{Unif})$ is one segment's expected log-likelihood ratio and $aL = (1 + o(1)) \log N$.*

(c) χ^2 ***regime.*** *If $K'(2) \leq a/2 + K(2)/2$, then $D(K, a) = a/2 + K(2)/2$, where $K(2)L = (1 + o(1)) \sum_{t \in block} \log(1 + \chi^2(F_{P_t} \| \mathrm{Unif}))$.*

(d) ***Strict gain.*** *$D(K, a) > a/2$. If moreover $P_t \equiv P_n$ for all tokens, order-blind tests fail for every $p > a/2$ (Proposition 5.1).*

Intuition. (b) Full detectability is a Bonferroni criterion: a single segment must stand out against the N places it could be, so its expected evidence must exceed $\log N$; this holds for every NTP. (c) With little evidence per segment, a second-moment calculation sets the boundary: each watermarked block contributes χ^2-information $n^{K(2)}$, buying exactly $K(2)/2$ beyond the i.i.d. boundary (for Gaussian blocks this is the Donoho–Jin line $p = 1/2 + r$). (d) Any positive evidence per segment moves the boundary past the line order-blind tests cannot cross. Between the two universal ends, the shape of K matters (Figure 2): at

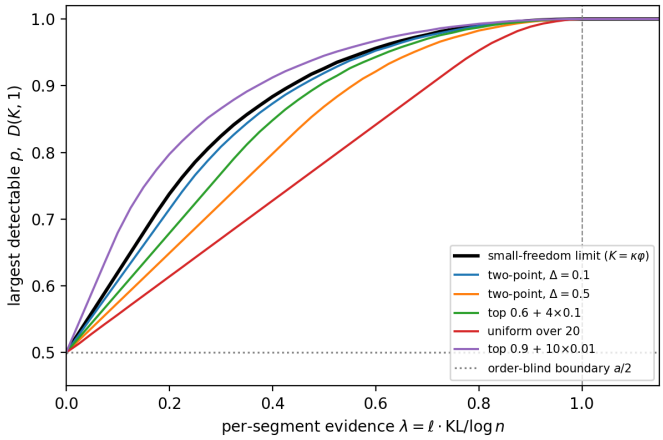


Figure 2: Sharp boundary $D(K,1)$ for fixed NTPs against the per-segment evidence $\lambda = \ell \cdot \mathrm{KL}(F_P \| \mathrm{Unif})/\log n$. The ends are universal: the order-blind boundary $1/2$ at $\lambda = 0$, and saturation exactly at $\lambda = a = 1$. The black curve is the small-freedom limit $K = \kappa\varphi$.

equal per-segment KL, the boundaries of different NTPs differ by up to about 0.2, and evidence concentrated in rare, near-certain tokens (a peaked top choice with a tail of unlikely alternatives) is easier to detect than evidence spread evenly (a flat NTP). This is the HC phenomenon of Lecture 3 in another guise: a statistic that looks for a few extreme blocks gains when per-block evidence is heavy-tailed.

Small freedom: events per segment, and a key-only test. In the small-freedom instance the boundary depends only on κ, the expected number of events per segment divided by $\log n$.

Corollary 5.5 (Events per segment)**.** *Let $P_t \equiv P^{\Delta_n}$, $\Delta_n = n^{-q}$, $\ell_n \Delta_n = \kappa \log n$ and $a = 1 - q$, so $K = \kappa\varphi$; write $D_{\mathrm{full}}(\kappa, a) = D(\kappa\varphi, a)$. Then:*

(i) D_{full} is nondecreasing and lower semicontinuous in κ;

(ii) if $\kappa \le \kappa_0(a) \equiv a/(2\varphi'(2) - \frac{1}{2})$, then $D_{\mathrm{full}} = a/2 + \kappa/4$;

(iii) $D_{\mathrm{full}} \ge a$ if and only if $\kappa\varphi'(1) \ge a$;

(iv) (key-only test) for fixed $u > 0$, among tests based on the block counts $C_k(u) = \#\{t \in$ block $k : Y_t > 1 - u\Delta_n\}$, the boundary is $D(K_u, a)$, with K_u the counts' cumulant (Lemma B.2), attained by ψ^* on these counts. Let $D_{\mathrm{cnt}}(\kappa, a) = \sup_u D(K_u, a) \le D_{\mathrm{full}}(\kappa, a)$.

Intuition. Discarding tokens shortens blocks, so more events never hurt (i); (ii)–(iii) are Theorem 5.4(c) and (b) with $K(2) = \kappa/2$ and $K'(1) = \kappa\varphi'(1)$. The key-only test sees only approximately Poisson counts, so it pays for discarding where in the block each large pivot fell. It uses the key and a threshold $1 - u\Delta_n$ but never the NTPs; scanning a fine geometric grid of thresholds and block lengths removes the need to know Δ_n and ℓ_n (Remark E.2). Table 2 gives numbers at $a = 0.7$: the key-only test loses little (0.546 vs. 0.576 at $\kappa = 1$), while every order-blind test is stuck at 0.35.

κ	0.2	0.5	1	1.5	2	3	4
D_{full} (oracle)	0.400	0.475	0.576	0.634	0.669	0.698	0.700
D_{cnt} (key-only)	0.391	0.452	0.546	0.607	0.647	0.689	0.700
order-blind	0.350	0.350	0.350	0.350	0.350	0.350	0.350

Table 2: Largest detectable p in the small-freedom limit at $a = 0.7$ ($\kappa_0 = 0.588$, saturation at $\kappa^* = 3.353$). Achievability and impossibility exponents were computed independently and agree to grid precision.

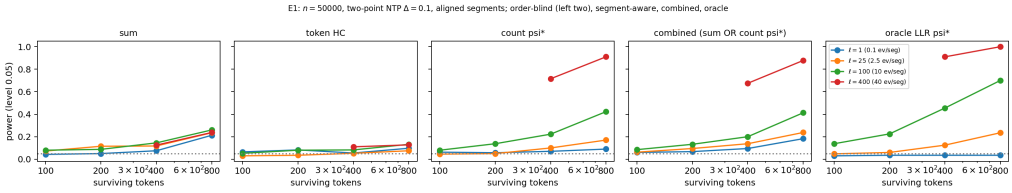


Figure 3: E1: power vs. number of surviving tokens, for four segment lengths.

Unknown positions and lengths. The alignment in Assumption A6 is a convenience: running ψ^* on a few shifted grids for each block length in a geometric grid, with a Bonferroni correction, attains every $p < D(K, a)$ when segments are arbitrary intervals of unknown length $\geq \ell_n$ (Proposition E.1 in Appendix E).

6 Experiments

Setup. All experiments simulate pivots exactly under Assumption A1, with critical values calibrated by Monte Carlo under H_0 at level 0.05 (300–400 replications; standard error ≤ 0.03). We compare *order-blind* tests (the class sum test $\sum_t S_t$, and token-level HC on $1 - Y_t$, the HC member of the Tr-GoF family); *segment-aware* tests (ψ^* on key-only block counts above $1 - 1.3\Delta$ with randomized binomial p-values, or on oracle block log-likelihood ratios with Monte Carlo p-values); and the *combined* test, which rejects if the sum test or the count-based ψ^* does, with Bonferroni-split levels. Code is in `code/` (see its README).

E1: segment length at fixed survival. With $n = 5 \times 10^4$, $P_t \equiv P^{0.1}$ and segment lengths $\ell \in \{1, 25, 100, 400\}$ (0.1 to 40 expected events per segment), we vary the number of surviving tokens (Figure 3). Order-blind tests are flat in ℓ (power ≤ 0.26), as Proposition 5.1 predicts, while segment-aware tests improve sharply with events per segment: with 800 survivors the oracle has power $0.04, 0.23, 0.70, 1.00$ and the key-only test $0.09, 0.17, 0.42, 0.91$ for $\ell = 1, 25, 100, 400$. The combined test stays within 0.03 of the better of its components.

E2: finite-n phase diagram. With $n = 2^{18}$ and blocks of length 512 ($N = 512$, $a = 1/2$), $\Delta = \kappa \log n/\ell$ and $\text{round}(n^{-p}N)$ aligned segments (Figure 4), token HC detects only near the order-blind boundary $p = a/2$, while segment-aware power extends far beyond it and grows with κ: at $\kappa \geq 2$ the oracle keeps power 0.69–0.96 even at $p = 0.45$ (two signal segments),

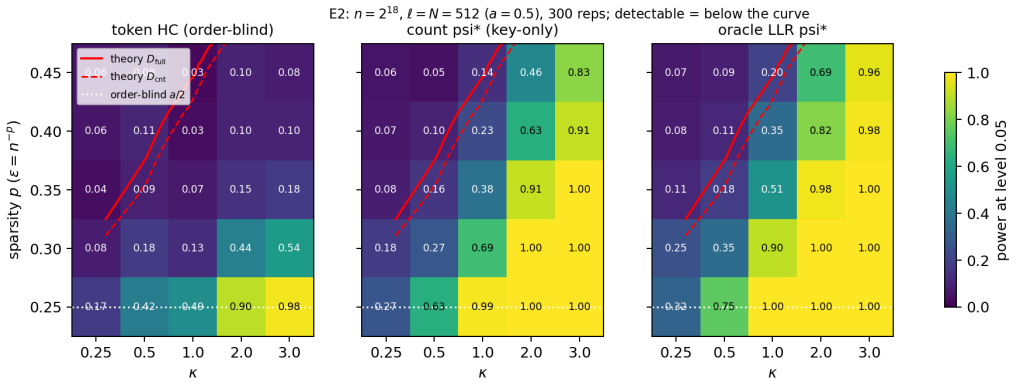


Figure 4: E2: empirical power over (p, κ) with the theoretical boundaries (red: D_{full} solid, D_{cnt} dashed; white dotted: order-blind $a/2$).

where token HC has power ≤ 0.10. The empirical 50%-power contour ($p \approx 0.27, 0.35, 0.45$ at $\kappa = 0.5, 1, 2$) lies below the asymptotic curve $(0.37, 0.44, 0.50)$, the usual slow convergence of log-scale asymptotics at $\log n \approx 12.5$, but its shape and its separation from the order-blind boundary match the theory.

E3–E4: outside the sandbox. (Details and figures in Appendix F.) With segments of length 100 placed at random, a single grid loses substantial power (0.53 vs. 0.84 aligned), while four shifted grids (0.68) or a fully length-adaptive scan (0.71) recover most of it, as Proposition E.1 predicts. With heterogeneous NTPs (60% near-deterministic tokens) and unknown Δ, the key-only test scanning four thresholds rises from power 0.08 to 1.00 as ℓ goes from 1 to 400, while order-blind tests stay between 0.15 and 0.40; under i.i.d. survival the sum test is better, and the combined test is a good default across settings.

E5: essay-length text. At $n = 2000$ the theory's criterion is concrete: with sentence blocks ($\ell = 25$, $N = 80$), saturation needs per-segment evidence $\log N \approx 4.4$ nats. At $\Delta = 0.4$ one sentence carries about this much, and four surviving sentences (5% of the text) give the oracle block test power 0.89 and the key-only test 0.62, against 0.29 for the sum test. At $\Delta = 0.1$ a sentence carries only 0.8 nats and the block tests gain nothing at $\ell = 25$; the gain appears only for paragraph-length segments (Appendix F).

7 Discussion

Summary. Contiguity is worth exactly as much evidence as a segment accumulates. When segments carry little evidence, nothing beats the i.i.d.-optimal order-blind tests; once they carry enough, those tests become strictly suboptimal and HC across blocks is optimal throughout the transition, which is complete when one segment's expected log-likelihood ratio exceeds $\log N$. A key-only count test captures most of the gain, and combining it with an order-blind test costs little.

Practical reading. The many-segment regime of our sharp theory fits sentence-level partial retention: a writer keeps a modest fraction of AI sentences verbatim, giving many short runs. The few-passages case (pasting two or three AI paragraphs into an essay) is the endpoint $p \to a$, where the answer is the saturation criterion: a passage is detectable once its expected log-likelihood ratio exceeds $\log N$. Low-entropy content (code, factual answers) carries little evidence per token, so it needs much longer runs, and there no test can exploit contiguity (Theorem 5.2(i)).

Limitations.

- *Pivot model (A1).* Real NTPs depend on the edited text, and repeated contexts repeat keys, so the null is uniform only after deduplication; we inherit these idealizations from Li et al.
- *Segments (A2).* We posit runs rather than model the editing process, and we detect but do not localize. The model fits copy-paste and sentence-level rewriting, not light scattered edits, which do not create runs.
- *NTPs (A3, A6).* The sharp results assume the same, known NTPs in every block. Which NTPs are hardest for the block problem is open: at equal evidence the boundary varies with the NTP (Figure 2), so the two-point law need not be least favorable here.
- *Asymptotics (A4).* Log-scale boundaries converge slowly (E2) and say nothing about calibration at the very small false-positive rates used in practice; our experiments calibrate by simulation.
- *Pivot-only tests.* Text-likelihood evidence (perplexity-style detectors) could be combined with pivots, at the cost of the exact, model-free null.
- *Synthetic data and simulation scale.* All experiments simulate pivots under the sandbox. E1–E4 use $n = 5 \times 10^4$ to 2.6×10^5 tokens (a thesis or a book, or pooled text), far longer than a single essay or article; E5 checks an essay-length $n = 2000$. The NTPs are illustrative rather than measured: $\Delta = 0.1$ (0.03 nats per token) is closer to low-temperature or code-like text, and open-ended prose likely carries more evidence per token.

Future work. The most immediate next step is a real-data study with an open model (e.g., GPT-2) and our own Gumbel-max key, using human text from C4 (RealNewsLike), the course's MADStat abstracts, and code as a low-entropy control: (i) measure the per-token KL across genres, i.e., how much evidence a real sentence or paragraph carries, which places real text on the axes of Figure 2; (ii) a copy-paste demo splicing watermarked passages into human documents; and (iii) sentence-level partial retention, keeping a random fraction of AI-rewritten sentences. Edits would be simulated by span replacement, but the model, text and NTPs would be real. Further directions are localizing segments rather than only detecting them, extending the theory to green/red-list watermarks (whose per-token signal is weak and dense, closer to the Gaussian template), and characterizing the least favorable NTPs for the block problem.

A Tools from sparse-mixture detection

Lower bounds: second moments and Hellinger distance. The impossibility side of a phase diagram (the "undetectable" region of Lecture 3, which we stated but did not prove in class) comes from Ingster (1997) and Donoho and Jin (2004). The idea is to place a random prior on where the signals are and show that the resulting mixture is statistically indistinguishable from the null. Two standard ways to show this are:

- the *second-moment method*: if the likelihood ratio Λ has $\mathbb{E}_0\Lambda^2 \to 1$, then total variation $\to 0$ and no test works;
- a *Hellinger-distance* bound: Hellinger distance adds over independent coordinates, so it suffices to bound the distance contributed by one coordinate.

The second moment is cleanest when the signal is not too strong. The Hellinger route, combined with truncation, handles strong but rare signals.

General sparse mixtures. Donoho–Jin's boundary is for Gaussian signals. Cai and Wu (2014) characterize the boundary for an arbitrary null Q and alternative G in the mixture $(1-\epsilon)Q + \epsilon G$. The boundary is determined by one function: the log-likelihood ratio $\log \frac{dG}{dQ}$ evaluated at the null quantile with tail probability N^{-s}, as a function of s. They also show that HC attains it adaptively. We need this generalization because our "observations" will be blocks of tokens, which are far from Gaussian.

Large deviations on the log scale. In Lecture 3 we wrote $\bar{\Phi}(\sqrt{2q\log p}) = L_p p^{-q}$: only the polynomial order of a tail probability matters. Large-deviations theory generalizes this to sums of many small terms. If T is a sum whose log moment-generating function satisfies $\log \mathbb{E}e^{\theta T} \approx K(\theta)\log n$, then the Cramér and Gärtner–Ellis theorems (Dembo and Zeitouni, Ch. 2) give

$$\mathbb{P}(T > t\log n) = n^{-I(t)+o(1)}, \qquad I(t) = \sup_\theta\{\theta t - K(\theta)\}.$$

The rate I is the Legendre transform of the cumulant K. Intuitively, the cheapest way for T to be large is to "tilt" each summand, and I is the cost of the best tilt. We use this to compute the tail of a block's evidence on the $\log n$ scale.

B Pivot lemmas and proofs for Sections 2 and 4

Two further facts about pivots are used for the count tests: how much mass a watermarked token puts just below 1, and the law of a block's count of large pivots.

Lemma B.1 (Excess tail mass)**.** *(a) If $P \in \mathcal{P}_\Delta$, $\Delta \le 1/2$ and $0 < s \le \Delta$, then $\mathbb{P}_P(Y > 1-s) \ge 2(1-e^{-1})s > 1.26\,s$, while $\mathbb{P}_{\mathrm{Unif}}(Y > 1-s) = s$.*

(b) For P^Δ and fixed $u > 0$: $r_u(\Delta) \equiv \mathbb{P}_{P^\Delta}(Y > 1-u\Delta) = \Delta\rho_u + O(\Delta^2)$, where $\rho_u = u + 1 - e^{-u}$.

Intuition. In a window of width $s \le \Delta$ just below 1, a watermarked token puts at least 1.26 times the uniform mass, however its freedom Δ is spread over alternatives. For the two-point

law, the excess $\Delta(1-e^{-u})$ saturates at Δ once the window is a few multiples of Δ wide. A wider window only adds null noise, which is why the best counting threshold sits near $1-1.3\Delta$.

Lemma B.2 (Block counts)**.** *For $u>0$ and $s=u\Delta$, let $C=\#\{t\le\ell : Y_t>1-s\}$. Then $C\sim\mathrm{Bin}(\ell,s)$ under H_0, and $C\sim\mathrm{Bin}(\ell,r_u(\Delta))$ for ℓ tokens with $P_t\equiv P^{\Delta}$. If $\ell\Delta=\kappa L$, the count's log-likelihood ratio T_u satisfies $\log\mathbb{E}_0 e^{\theta T_u}=L\big(K_u(\theta)+o(1)\big)$, where*

$$K_u(\theta)=\kappa\Big[u\big((\rho_u/u)^{\theta}-1\big)-\theta(\rho_u-u)\Big].$$

Intuition. A block's count of large pivots is approximately Poisson, with mean κuL under the null and $\kappa\rho_u L$ under the watermark. K_u is the corresponding Poisson cumulant on the $\log n$ scale.

Proof of Lemma 2.1. Let $E_w=-\log U_{t,w}\sim\mathrm{Exp}(1)$, so $w_t=\mathrm{argmin}_w E_w/P_{t,w}$. The minimum M of independent exponentials with rates $P_{t,w}$ is $\mathrm{Exp}(1)$ and is independent of the argmin, which equals w with probability $P_{t,w}$. Hence $Y_t=e^{-P_{t,w_t}M}=V^{P_{t,w_t}}$ with $V=e^{-M}\sim\mathrm{Unif}(0,1)$. For a token chosen without the key, the key is independent of the token, so $Y_t\sim\mathrm{Unif}(0,1)$. □

Proof of Lemma B.1. (a) $1-F_P(1-s)=\sum_w P_w\big(1-(1-s)^{1/P_w}\big)\ge\sum_w P_w(1-e^{-s/P_w})\ge(1-e^{-1})\sum_w\min(s,P_w)$, using $1-e^{-x}\ge(1-e^{-1})\min(x,1)$. It remains to show $\sum_w\min(s,P_w)\ge 2s$. If some $P_{w_0}\ge s$, that token contributes s; the others have total mass $1-P_{w_0}\ge\Delta\ge s$, so by subadditivity of $x\mapsto\min(s,x)$ they contribute $\ge s$. Otherwise every term equals P_w, and the sum is $1\ge 2s$. (b) $r_u(\Delta)=1-(1-\Delta)(1-u\Delta)^{1/(1-\Delta)}-\Delta(1-u\Delta)^{1/\Delta}$, with $(1-u\Delta)^{1/(1-\Delta)}=1-u\Delta+O(\Delta^2)$ and $(1-u\Delta)^{1/\Delta}=e^{-u}+O(\Delta)$. □

Proof of Lemma 4.1. Substituting $y=e^{-x}$ and factoring $e^{-\alpha x}$,

$$m_\Delta(\theta)-1=\underbrace{\int_0^\infty e^{-x}(e^{-\alpha\theta x}-1)\,dx}_{=-\alpha\theta/(1+\alpha\theta)}+\int_0^\infty e^{-(1+\alpha\theta)x}\big[(1+e^{-(\beta-\alpha)x})^{\theta}-1\big]\,dx.$$

The first term is $-\theta\Delta+O(\Delta^2)$. In the second, set $v=(\beta-\alpha)x$; then $1/(\beta-\alpha)=\Delta(1-\Delta)/(1-2\Delta)=\Delta+O(\Delta^2)$ and $c\equiv(1+\alpha\theta)/(\beta-\alpha)=O(\Delta)$. Since $|e^{-cv}-1|\le cv$ and $|(1+e^{-v})^{\theta}-1|\le|\theta|\max(1,2^{\theta-1})e^{-v}$, the second term equals $\Delta\int_0^\infty[(1+e^{-v})^{\theta}-1]\,dv+O(\Delta^2)$. Substituting $w=e^{-v}$ identifies the integral with $\varphi(\theta)+\theta$. The constants are uniform over compact sets (for negative θ we need $1+\alpha\theta>0$, true for small Δ). Then $\chi^2=m_\Delta(2)-1$, and $\mathrm{KL}(F_\Delta\|\mathrm{Unif})=m'_\Delta(1)$. Uniform convergence of the convex functions $(m_\Delta-1)/\Delta$ to the differentiable φ implies convergence of their derivatives, which gives the KL statement. The derivative values are elementary integrals, and $\varphi''(\theta)=\int_0^1(1+w)^{\theta}\log^2(1+w)\,w^{-1}dw>0$. □

Proof of Lemma B.2. $T_u=\lambda C+\ell\log\frac{1-r}{1-s}$ with $\lambda=\log\frac{r(1-s)}{s(1-r)}=\log(\rho_u/u)+O(\Delta)$. Then $\log\mathbb{E}_0 e^{\theta T_u}=\ell\log(1-s+se^{\theta\lambda})+\theta\ell\log\frac{1-r}{1-s}=\ell s(e^{\theta\lambda}-1)-\theta\ell(r-s)+O(\ell\Delta^2)$, using Lemma B.1(b). Substitute $\ell s=u\kappa L$ and $\ell(r-s)=(\rho_u-u)\kappa L+O(L\Delta)$. K_u is smooth and strictly convex, with $K_u(0)=K_u(1)=0$. □

C Proofs of Proposition 5.1 and Theorem 5.2

Proof of Proposition 5.1. Fix $S \in \mathcal{S}_n(\epsilon, \ell)$ with $|S| = k = \ell\lceil \epsilon n/\ell \rceil = (1 + o(1))\epsilon n$, a union of runs of length ℓ. By invariance and Assumption A1, $\mathbb{P}_S(\psi = 0) = \mathbb{P}_{\Pi S}(\psi = 0)$ for every permutation Π, so it equals the type II error under the prior π_k that picks a uniformly random k-subset. Hence $R_n(\psi) \geq 1 - \mathrm{TV}(\mathbb{P}_0, \mathbb{P}_{\pi_k})$. Write $\chi^2 = \chi^2(F_{P_n} \| \mathrm{Unif})$. The likelihood ratio $\Lambda = \mathbb{E}_{\pi_k} \prod_{t \in S} f_{P_n}(Y_t)$ satisfies $\mathbb{E}_0 \Lambda^2 = \mathbb{E}(1 + \chi^2)^H$, where $H = |S_1 \cap S_2|$ for independent $S_1, S_2 \sim \pi_k$ is hypergeometric. By Hoeffding (1963, Thm. 4), $\mathbb{E}g(H) \leq \mathbb{E}g(\mathrm{Bin}(k, k/n))$ for convex g. So $\mathbb{E}_0 \Lambda^2 \leq (1 + k\chi^2/n)^k \leq \exp(k^2\chi^2/n) = \exp\big((1 + o(1))n\epsilon^2\chi^2\big) \to 1$, and $\mathrm{TV} \leq \frac{1}{2}(\mathbb{E}_0 \Lambda^2 - 1)^{1/2} \to 0$. □

Proof of Theorem 5.2. (i) Let π mark each block of the aligned grid of length ℓ_n independently with probability $\epsilon' = 2\epsilon_n$. Then $|S| = \ell_n \mathrm{Bin}(N, \epsilon')$, and by Chernoff $\pi(\mathcal{S}_n(\epsilon, \ell)) \to 1$ because $N\epsilon_n = n^{1-p-\gamma+o(1)} \to \infty$. For any ψ,

$$R_n(\psi) \geq \mathbb{P}_0(\psi = 1) + \mathbb{P}_\pi(\psi = 0) - \pi(\mathcal{S}_n^c) \geq 1 - \mathrm{TV}(\mathbb{P}_0, \mathbb{P}_\pi) - o(1).$$

The likelihood ratio is $\prod_k (1 - \epsilon' + \epsilon' W_k)$, with independent block ratios satisfying $\mathbb{E}_0 W_k^2 = (1 + \chi^2)^\ell$. Hence

$$\mathbb{E}_0\Big[\prod_k (1 - \epsilon' + \epsilon' W_k)^2\Big] = \big(1 + \epsilon'^2[(1+\chi^2)^\ell - 1]\big)^N \leq \exp\big(4N\epsilon^2(e^{\ell\chi^2} - 1)\big) \leq \exp\big(Cn\epsilon^2\Delta\big) \to 1,$$

using $\ell\chi^2 = O(1)$ and $e^x - 1 \leq C'x$ on bounded sets. (ii) Under H_0 the count is $\mathrm{Bin}(n, \Delta)$, so by Chebyshev the type I error is $\leq L^{-2}$. Under H_1 it is a sum of independent indicators which, by Lemma B.1(a), stochastically dominates $B = \mathrm{Bin}(n - |S|, \Delta) + \mathrm{Bin}(|S|, 1.26\Delta)$. Here $\mathbb{E}B - n\Delta = 0.26|S|\Delta$ and $\mathrm{Var}\, B \leq 2n\Delta$. Since $|S|\Delta/(\sqrt{n\Delta}L) \geq n^{(1-2p-q)/2}/L \to \infty$, Chebyshev applied to B gives type II error $\to 0$. (iii) Every interval of length $\geq \ell_n$ contains a full block. Under H_0, $C_k \sim \mathrm{Bin}(b, \Delta)$. Bernstein's inequality with $x = \sqrt{3\lambda L} = o(\lambda)$ gives $\mathbb{P}(C_k > \lambda + x) \leq \exp\big(-\frac{x^2}{2(\lambda + x/3)}\big) = n^{-3/2+o(1)}$, and a union bound over $\leq n$ blocks controls the type I error. A full signal block has $C_k \succeq \mathrm{Bin}(b, 1.26\Delta)$. Its mean excess 0.26λ dominates both x and its standard deviation, since $\lambda \asymp n^{\gamma-q} \gg L$. □

Adaptivity. Without knowledge of (Δ_n, ℓ_n), apply (ii) over thresholds $s \in \{2^{-j}\}$ and (iii) over $(s, b) \in \{2^{-j}\} \times \{2^j\}$, with Bonferroni corrections. Some grid point has $s \in [\Delta/2, \Delta]$ and $b \in [\ell/4, \ell/2]$, where the same arguments go through with constants halved. The type I errors sum to $O(\log n/L^2) + O(\log^2 n \cdot n^{-1/2+o(1)}) \to 0$.

D Proof of Theorem 5.3

Remark D.1 (Relation to known results)**.** After the change of units $s = I(t)/a$, $\varsigma(s) = t/a$, $\beta = p/a$, the condition $p < D(K, a)$ is exactly Cai and Wu's (2014) boundary $\beta < \frac{1}{2} + \sup_s\{\varsigma(s) - s + \frac{1}{2}(s \wedge 1)\}$. Their assumptions are stated in terms of log-likelihood ratios at null quantiles; ours, convergence of block cumulants, are immediate for sums over tokens. For Gaussian blocks, $K(\theta) = r\theta(\theta - 1)$, and with $a = 1$ the condition reduces to $r > \rho_{\mathrm{detect}}(p)$, the Donoho–Jin boundary of Lecture 3 (checked numerically). So Theorem 5.3 contains the class result as a special case.

Throughout, $I_1(t) \equiv \sup_\theta\{\theta t - K(\theta+1)\} = I(t) - t$ is the rate function under G_n, since $\log \mathbb{E}_{G_n} e^{\theta T} = \Lambda_n(\theta+1)$. Define

$$g(t) = a - I_1(t) - \max\left\{0, \tfrac{a-I(t)}{2}\right\},$$

so that $D(K,a) = \sup_{t\geq t_0} g(t)$. In the counting heuristic, $g(t) > p$ says that the excess count at threshold tL beats both 1 and the null standard deviation.

(a) Lower bound. Let π mark blocks independently with probability $\epsilon' = 2\epsilon$. Then $\pi(|J| \geq \epsilon N) \to 1$, and as above $\inf_\psi R_n \geq 1 - \mathrm{TV}(\mathbb{P}_0, \mathbb{P}_\pi) - o(1)$. With $H^2(P,Q) = \int(\sqrt{dP} - \sqrt{dQ})^2$, we have $\mathrm{TV} \leq H$ and $H^2(\otimes P_k, \otimes Q_k) \leq \sum_k H^2(P_k, Q_k)$. Here $\mathbb{P}_\pi = M^{\otimes N}$ with $M = (1-\epsilon')Q_n + \epsilon' G_n$, and $H^2(Q_n, M) = \mathbb{E}_Q[(\sqrt{1+x}-1)^2]$ with $x = \epsilon'(e^T - 1) \geq -1$. We use $(\sqrt{1+x}-1)^2 \leq \min(x^2, |x|)$. On $\{T \leq y\}$, with $y = \log(1/\epsilon')$, use $x^2 \leq \epsilon'^2(e^{2T}+1)$. On $\{T > y\}$, use $|x| \leq \epsilon' e^T$ together with $\mathbb{E}_Q[e^T 1\{T > y\}] = G_n(T > y)$. Hence

$$NH^2(Q_n, M) \leq N\epsilon'^2 + N\epsilon'^2\, \mathbb{E}_Q[e^{2T} 1\{T \leq y\}] + N\epsilon'\, G_n(T > y).$$

For $\theta \in [0,2]$, $e^{2T}1\{T \leq y\} \leq e^{\theta T + (2-\theta)y}$, and for $\theta \geq 0$, $1\{T > y\} \leq e^{\theta(T-y)}$. With $y = pL - \log 2$, and Assumption A5 applied at a θ nearly attaining each infimum (if $p > t_+$ then $E_3 = -\infty$), the three terms are at most $n^{E_j + o(1)}$, where

$$E_1 = a - 2p, \qquad E_2 = a - 2p + \inf_{\theta\in[0,2]}\{K(\theta) + (2-\theta)p\}, \qquad E_3 = a - p - I_1^+(p),$$

with $I_1^+(p) = \sup_{\theta\geq 0}\{\theta p - K(\theta+1)\}$. If $p > D$, Lemma D.2 gives $\max_j E_j < 0$, so $\mathrm{TV} \to 0$.

Lemma D.2 (Duality)**.** *If* $\max\{E_1, E_2, E_3\} \geq 0$*, then* $p \leq D(K,a)$*.*

Proof. We exhibit $t \geq t_0$ with $g(t) \geq p$.

- $E_1 \geq 0$: take $t = t_1$, where $I_1 = 0$ and $I = t_1$; then $g(t_1) = \min\{a, (a+t_1)/2\} > a/2 \geq p$.
- $E_2 \geq 0$: let $\tilde{I}(t) = \sup_{\theta\in[0,2]}\{\theta t - K(\theta)\}$, so that $2t - \tilde{I}(t) = \inf_{\theta\in[0,2]}\{K(\theta) + (2-\theta)t\}$. For each $\theta \in [0,2]$ the bracket is nondecreasing in t, hence so is the infimum, and $\inf_{\theta\in[0,2]}\{K(\theta)+(2-\theta)p\} = \sup_{t\leq p}\{2t - \tilde{I}(t)\}$. The map $t \mapsto 2t - \tilde{I}(t)$ is nondecreasing (slope $2 - \theta(t) \geq 0$), so the supremum is attained at $t^\star = \min\{p, K'(2)\} \in (0, K'(2)]$, where $\tilde{I}(t^\star) = I(t^\star)$. Thus $a - 2p + 2t^\star - I(t^\star) \geq 0$. Then $a - p - I_1(t^\star) = a - p - I(t^\star) + t^\star \geq p - t^\star \geq 0$, and $a - p - I_1(t^\star) - \frac{a-I(t^\star)}{2} = \frac{1}{2}(a - I(t^\star) + 2t^\star - 2p) \geq 0$; so $g(t^\star) \geq p$.
- $E_3 \geq 0$ and $p \geq t_1$: then $I_1^+(p) = I_1(p)$ and $I(p) = I_1(p) + p$. With $x = a - p - I_1(p) \geq 0$, we have $\frac{a-I(p)}{2} = \frac{x}{2} \leq x$, so $g(p) \geq p$.
- $E_3 \geq 0$ and $p < t_1$: then $g(t_1) = \min\{a, (a+t_1)/2\} > p$, because $p < \min\{a, t_1\}$.

□

Lemma D.3. *g is continuous on (t_0, t_+), equals $-\infty$ on (t_+, ∞), and is strictly increasing on $[t_0, 0]$. Hence $D(K,a) = \sup_{t>0} g(t)$, and if $p < D$ there is $t > 0$ with $g(t) > p$ and $I(t) \neq a$.*

Proof. I is convex and finite on (t_0, t_+), hence continuous there, and so is g; on (t_+, ∞), $I = +\infty$ and $g = -\infty$. Write $g(t) = a - I(t) + t - \max\{0, (a - I(t))/2\}$, and note $I'(t) = \theta(t)$ with $K'(\theta(t)) = t$. The slope of g is $1 - \theta(t)$ where $I(t) > a$ and $1 - \theta(t)/2$ where $I(t) < a$. On $[t_0, 0]$ we have $0 \le \theta(t) \le \theta(0) < 1$, because $K'(1) = t_1 > 0$. So both slopes are positive and g is strictly increasing there. Since I is strictly increasing on (t_0, t_+), a t with $g(t) > p$ can be taken in $(0, t_+)$, where $I(t) < \infty$ (if $t = t_+ < \infty$, move t slightly left, using continuity of the closed convex I on its domain), and the single point with $I(t) = a$ can be avoided by continuity. □

(b) Upper bound. By Assumption A5, T/L satisfies the Gärtner–Ellis hypotheses under Q_n with speed L: the limit K is finite and differentiable on the open interval $(\theta_-, \infty) \ni 0$ and, if $\theta_- > -\infty$, blows up at θ_- (so $K' \to -\infty$ there, by convexity); hence it is essentially smooth (Dembo and Zeitouni, Thm. 2.3.6). Gärtner–Ellis also needs the limit for every θ: for $\theta \le \theta_- < \theta' < \theta''$, convexity gives $\Lambda_n(\theta) \ge \Lambda_n(\theta') + (\theta' - \theta)\frac{\Lambda_n(\theta') - \Lambda_n(\theta'')}{\theta'' - \theta'}$, and dividing by L and letting $\theta' \downarrow \theta_-$ shows $\Lambda_n(\theta)/L \to +\infty$. So the limit is K extended by $+\infty$, which is lower semicontinuous because $K \to \infty$ at θ_-. This gives a large deviation principle with rate I; under G_n the limit is $K(\cdot + 1)$, with rate I_1. Hence $h_0(t) \equiv Q_n(T \ge tL) = n^{-I(t)+o(1)}$ for $t > t_0$. Likewise $h_1(t) \equiv G_n(T \ge tL) = n^{-I_1(t)+o(1)}$ for $t > t_1$ and $h_1(t) \to 1$ for $t < t_1$; in all cases $h_1 \ge n^{-I_1(t)+o(1)}$. By Lemma D.3, fix $t > 0$ with $g(t) > p$ and $I(t) \ne a$. Note that $\{T_k \ge tL\} \subseteq \{\pi_k \le h_0\}$, since $\pi_k \le Q_n(T \ge T_k) \le Q_n(T \ge tL)$; and under H_0 the π_k are i.i.d. Unif$(0, 1)$, so null counts below any level α are exactly Bin$(\cdot, \alpha)$.

Case $I(t) < a$. Then $h_0 \gg 1/N$; choose a dyadic level $\alpha \in [h_0, 2h_0]$. The count $X_\alpha = \#\{k : \pi_k \le \alpha\}$ dominates Bin$(N - |J|, \alpha)$ + Bin$(|J|, h_1)$, so $X_\alpha - N\alpha \ge |J|(h_1 - \alpha) - O_P(\sqrt{N\alpha} + \sqrt{|J|h_1})$. Since $h_1/h_0 = n^{t+o(1)} \to \infty$ and $|J| \ge \epsilon N$,

$$\frac{|J|(h_1 - \alpha)}{\sqrt{N\alpha}} \ge n^{a-p-I_1(t)-(a-I(t))/2+o(1)} = n^{g(t)-p+o(1)} \gg L^2,$$

and $|J|h_1 \to \infty$ because $g(t) > p$ forces $a - p - I_1(t) > 0$.

Case $I(t) > a$. The number of signal blocks with $T_k \ge tL$ stochastically dominates Bin$(|J|, h_1)$, whose mean is $\ge n^{g(t)-p+o(1)} \to \infty$. So with probability $\to 1$ some block has $\pi_k \le h_0 = n^{-I(t)+o(1)} < (NL)^{-1}$.

Type I error. For each dyadic level $N\alpha \ge 1$, and Bernstein gives $\mathbb{P}_0\big(\text{Bin}(N, \alpha) - N\alpha > x\sqrt{N\alpha}\big) \le \exp\big(-\frac{x^2}{2(1+x/3)}\big) \le e^{-x/2}$ for $x \ge 3/2$. With $x = L^2$, a union bound over $\log_2 N$ levels gives $o(1)$, and $\mathbb{P}_0(\min_k \pi_k < (NL)^{-1}) \le 1/L$.

E Proofs of Theorem 5.4, Corollary 5.5 and Proposition E.1

Proof of Theorem 5.4. (a) Block k has likelihood ratio e^{T_k} between its watermarked law $\bigotimes_{t \in \text{block } k} F_{P_t}$ and its null law Unif$^{\otimes \ell}$, with $T_k = \sum_{t \in \text{block } k} \log f_{P_t}(Y_t)$. Hence $(T_k)_k$ is sufficient for the set J of watermarked blocks: total variation between the null and any mixture over J, and so the lower bound, can be computed on (T_k), and ψ^* is a function of (T_k). The T_k are independent, and since every block has the same multiset of NTPs, their null

and alternative laws do not depend on k (blocks with the NTPs in different orders have permuted product laws, but the same law of T_k). So Theorem 5.3 applies to (T_k). By independence (Assumption A1), $\Lambda_n(\theta) = \sum_{t\in\text{block}} \log m_{P_t}(\theta)$, which satisfies Assumption A5 by Assumption A6. Apply Theorem 5.3; the test ψ^* uses Q_n, which the oracle knows. (b) $g(t) \le a - I_1(t)$, and $I_1 \ge 0$ vanishes only at $t_1 = K'(1)$. So $\sup g \ge a$ forces $t = t_1$, where $g(t_1) = a - \max\{0, (a-t_1)/2\}$; this is $\ge a$ iff $t_1 \ge a$. The supremum is attained because $g(t) \to -\infty$ as $t \to \infty$. For the interpretation: Λ_n/L are convex and converge pointwise to K, which is differentiable at 1, so $\Lambda_n'(1)/L \to K'(1)$, and $\Lambda_n'(1) = \sum_{t\in\text{block}} \mathrm{KL}(F_{P_t}\|\mathrm{Unif})$; also $\log N = (a + o(1))L$. (c) Let $p \in [K'(2),\, a/2 + K(2)/2]$, a nonempty interval by assumption. Since $K'(\theta) \le p$ on $[0,2]$, $\inf_{\theta\in[0,2]}\{K(\theta) + (2-\theta)p\} = K(2)$, so $E_2 = a - 2p + K(2) \ge 0$, and Lemma D.2 gives $p \le D$; thus $D \ge a/2 + K(2)/2$. Conversely, take any $p \in (a/2 + K(2)/2,\, a)$. Then $E_1 < 0$ and $E_2 \le a - 2p + K(2) < 0$. Also $p > K'(2) \ge t_1$, so $I_1^+(p) = I(p) - p$, and $I(p) \ge 2p - K(2) > a$ (take $\theta = 2$), hence $E_3 < 0$. By the proof of Theorem 5.3(a) the problem is undetectable, so $D \le p$. Finally $\Lambda_n(2) = \sum_{t\in\text{block}} \log(1 + \chi^2(F_{P_t}\|\mathrm{Unif}))$. (d) K is strictly convex with $K(0) = K(1) = 0$, so $t_1 = K'(1) > 0$ and $g(t_1) = a - \max\{0, (a-t_1)/2\} > a/2$. The statement about order-blind tests is Proposition 5.1: for fixed P, χ^2 is a constant and $n\epsilon^2 = n^{1-2p}$; for P^{Δ_n}, $\chi^2 \sim \Delta_n/2$ (Lemma 4.1) and $n\epsilon^2\Delta_n = n^{1-2p-q}$. □

Proof of Corollary 5.5. By Lemma 4.1, $\Lambda_n(\theta) = \ell \log m_\Delta(\theta) = \ell\Delta\varphi(\theta) + O(\ell\Delta^2) = L\kappa\varphi(\theta) + o(L)$ for every $\theta \in \mathbb{R}$ ($m_\Delta(\theta)$ is finite once $\theta > -1/\alpha$, and $\alpha \to 0$). The limit $K = \kappa\varphi$ is finite, differentiable and strictly convex on $\mathbb{R}$ with $K' \to \infty$, so Assumption A6 holds with $a = 1 - q$. (i) Monotonicity: discarding the last $(1 - \kappa/\kappa')\ell$ tokens of each block reduces a κ'-problem to a κ-problem with the same (a, p), since $\log n$ changes by $O(1)$. Lower semicontinuity: by Lemma D.3, $D(\kappa) = \sup_{t>0} g_\kappa(t)$, an index set not depending on κ. For fixed $t > 0$, $\kappa \mapsto I_\kappa(t) = \sup_\theta\{\theta t - \kappa\varphi(\theta)\}$ is convex and finite, hence continuous. So D is a supremum of continuous functions, and with monotonicity it is left-continuous. The lower-semicontinuity argument applies to $c \mapsto D(cK, a)$ for any K satisfying Assumption A5; monotonicity in c uses the token-discarding argument and hence needs all tokens to share one NTP, which holds in both instances of Assumption A6. (ii)–(iii) Apply Theorem 5.4(c) and (b) with $K(2) = \kappa\varphi(2) = \kappa/2$, $K'(2) = \kappa\varphi'(2)$ and $K'(1) = \kappa\varphi'(1)$; the condition $K'(2) \le a/2 + K(2)/2$ is $\kappa\varphi'(2) \le a/2 + \kappa/4$, i.e., $\kappa \le \kappa_0$, and then $D = a/2 + K(2)/2 = a/2 + \kappa/4$. (iv) The count vectors are i.i.d. across blocks under the prior, with the binomial laws of Lemma B.2; apply Theorem 5.3 to the reduced data. □

Proposition E.1 (Unknown alignment and segment length)**.** *In either instance of Assumption A6, let segments be arbitrary intervals of length $\ge \ell_n$, with ℓ_n unknown. Running ψ^* on k shifted grids for each block length in a geometric grid, with a Bonferroni correction, detects whenever $p < D\big((k/(k+1))^2 K, a\big)$, hence every $p < D(K, a)$ for k large. The lower bound of Theorem 5.4(a) still applies.*

Intuition. Some shifted grid fits a full block inside each segment, losing at most a $(k/(k+1))^2$ fraction of its evidence, and $O(k \log n)$ grids cost only a polylogarithmic multiplicity.

Proof of Proposition E.1. Some grid length ℓ' satisfies $\ell'(1 + 1/k) \le \ell_n \le \ell'(1 + 1/k)^2$. Block starts of the k shifted grids are spaced ℓ'/k apart, so each segment contains a full block of some grid. Counting full signal blocks over all grids gives $\gtrsim |S|/\ell'$, so one grid contains

$\gtrsim |S|/(k\ell')$ of them. Its signal-block fraction is $n^{-p+o(1)}$. In either instance all tokens share the NTP P_n, so a full signal block of length ℓ' has cumulant $c_n\Lambda_n$, with $c_n = \ell'/\ell_n \in [c_0, 1]$, $c_0 = (k/(k+1))^2$. The ratio c_n need not converge, so argue along subsequences: every subsequence has a further one with $c_n \to c \geq c_0$, along which the block cumulants converge to cK. Since all tokens share the NTP P_n, discarding tokens reduces a c-problem to a c_0-problem, so $D(cK, a) \geq D(c_0K, a) > p$ and ψ^* has risk $\to 0$ along it; hence $R_n \to 0$. Lower semicontinuity of $c \mapsto D(cK, a)$ at $c = 1$ gives the final claim. The remaining blocks of that grid may be partly watermarked rather than null. This only helps: $f_P(y) = \sum_w y^{1/P_w - 1}$ is increasing for every NTP P, so a watermarked pivot is larger than a uniform one in likelihood-ratio order, and T_k (a sum of the increasing functions $\log f_{P_t}$) is then stochastically larger on a partly watermarked block than on a null block. The same holds for the block counts, whose summands $1\{Y_t > 1 - u\Delta_n\}$ are increasing in Y_t. The counts entering $\mathrm{HC}_N^{\mathrm{dy}}$ can therefore only increase, and the lower bounds on them used in the proof of Theorem 5.3(b) still hold. There are $O(k^2 \log n)$ tests. Replacing $(NL)^{-1}$ by $(NL^3)^{-1}$ in ψ^* makes the Bonferroni sum of type I errors vanish without changing any exponent. □

Adaptivity of the key-only test.

Remark E.2 (What the key-only test needs)**.** At a fixed u, the count test uses the threshold $1 - u\Delta_n$, and hence the scale Δ_n, as well as the block length ℓ_n; it never uses the next-token probabilities themselves. The scale and length can be removed. Fix $\eta > 0$, scan thresholds $s \in \{(1+\eta)^{-j} : 1 \leq j \leq \log_{1+\eta} n\}$ together with the block-length grid of Proposition E.1, and apply a Bonferroni correction over these $O(k^2\eta^{-1}\log^2 n)$ tests (including the shifts), using the level $(NL^3)^{-1}$ of Proposition E.1; this does not change any exponent. Some grid threshold equals $u'\Delta_n$ with u' within a factor $1+\eta$ of any target u. The map $u \mapsto D(K_u, a)$ is lower semicontinuous (for fixed $t > 0$ the function g of Lemma D.3 depends continuously on u, and D is a supremum of such functions), so $\{u : D(K_u, a) > p\}$ is open. Hence, for η small enough, the scan detects every $p < D_{\mathrm{cnt}}(\kappa, a)$ (a routine compactness argument handles the dependence of u' on n, using that the $o(1)$ in Lemma B.2 is locally uniform in u, by its explicit formulas). With a factor-2 grid, as in the coarse adaptivity argument, it would instead be guaranteed only the worst case, over the position of u^* within a grid cell, of the better of the two neighboring grid points.

F Experiments E3–E5

E3: unknown alignment and length. Segments of length 100 are placed at random ($n = 5 \times 10^4$, $\Delta = 0.1$; Figure 5).

- Misalignment costs a single grid substantial power: 0.53 vs. 0.84 for the aligned oracle at 1600 survivors.
- Four shifted grids recover most of the loss (0.68), and a fully length-adaptive multiscale scan is as good (0.71), as Proposition E.1 predicts.

E4: heterogeneous NTPs, unknown Δ. This setting lies outside the sharp theory, which needs the same NTPs in every block; with i.i.d. random NTPs this holds only approx-

imately, for typical blocks. Each token is near-deterministic ($\Delta_t = 0.01$) with probability 0.6; otherwise $\Delta_t \sim \mathrm{Unif}(0.1, 0.5)$, with the residual mass split over $\{1, 2, 5, 20\}$ tokens (mean $\Delta_t \approx 0.13$). The key-only test scans four thresholds (Figure 5).

- With 800 survivors, its power rises from 0.08 to 0.25, 0.85 and 1.00 as ℓ goes from 1 to 25, 100 and 400; order-blind tests stay between 0.15 and 0.40.
- Under i.i.d. survival ($\ell = 1$) the sum test is better (0.29). The combined test gives 0.17 there and 1.00 at $\ell = 400$, which supports it as the practical default.

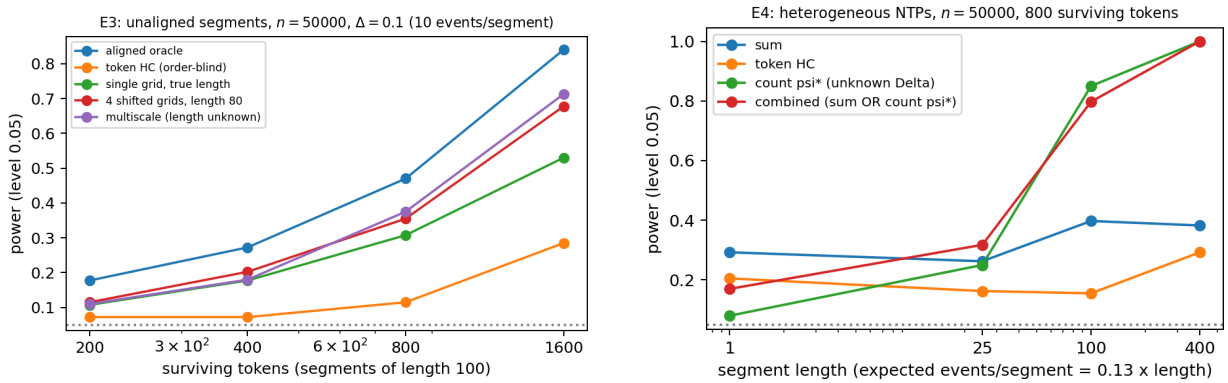


Figure 5: Left, E3: unaligned segments of unknown length. Right, E4: heterogeneous NTPs with unknown Δ_t.

E5: short documents. The design of E1 at an essay-like length, $n = 2000$, for $\ell \in \{1, 25, 100\}$ and two-point NTPs with $\Delta = 0.1$ (0.031 nats per token) and $\Delta = 0.4$ (0.17 nats per token); 2000 replications per setting (Figure 6). The prediction is that block tests gain over order-blind ones once a segment's evidence $\ell \cdot \mathrm{KL}$ is comparable to $\log N = \log(n/\ell)$.

- Order-blind tests are flat in ℓ, as in E1 (sum test: 0.33, 0.33, 0.28 at 200 survivors and $\Delta = 0.1$).
- $\Delta = 0.1$: a sentence carries $0.8 \ll \log 80$ nats, and at $\ell = 25$ the block tests are no better than the sum test (0.69 vs. 0.75 at 400 survivors). A paragraph carries $3.1 \approx \log 20$ nats, and at $\ell = 100$ the oracle block test gains (0.56 vs. 0.28 at 200 survivors; key-only 0.39).
- $\Delta = 0.4$: a sentence carries $4.4 \approx \log 80$ nats, and the gain is large: with 100 survivors (four sentences) the oracle and key-only block tests have power 0.89 and 0.62, against 0.29 for the sum test. One surviving paragraph is detected with power 0.98 by the key-only test, against 0.28.
- The combined test loses at most 0.17 in power to the better of the sum and key-only tests (worst under i.i.d. survival).

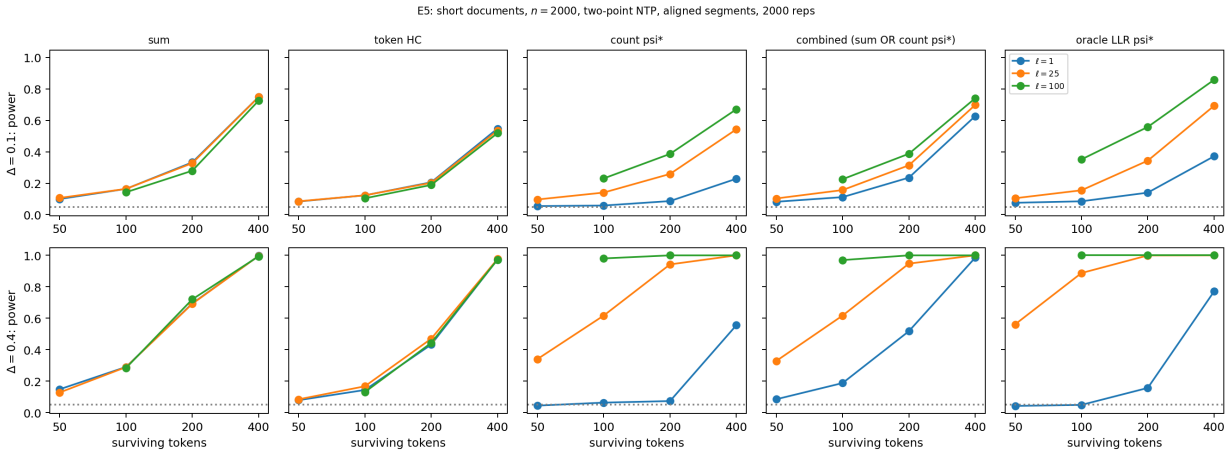


Figure 6: E5: power at $n = 2000$ against the number of surviving tokens; top row $\Delta = 0.1$, bottom row $\Delta = 0.4$.

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